

CYBERHOST: A ONE-STAGE DIFFUSION FRAMEWORK FOR AUDIO-DRIVEN TALKING BODY GENERATION

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01

CyberHost surpasses DiffTED on A2V-B

- FID 32.97 vs DiffTED 36–38 (lower is better)
- FVD 555.8 vs 612–644 (lower is better)
- CSIM 0.514; HKV 0.884 vs 0.42–0.47
- Same test split and inputs (single ref image + speech)

02

One-stage diffusion overcomes two-stage error compounding

- Two-stage audio→pose→video compounds errors and weakens audio–limb correlation
- Small critical regions (face, hands) underfit under audio-only supervision
- Our one-stage diffusion generates video directly from audio + one image
- Region Attention Modules fuse a latents bank with identity descriptors
- Human-prior conditions (movement map, pose alignment) stabilize motion

03

Region-aware diffusion and human priors enable stable, identity-faithful motion

- RAMs focus faces/hands via a spatio-temporal latents bank and identity descriptors.
- Human priors (movement map, hand-clarity, pose-aligned refs) stabilize global/local motion
- Ablation: removing RAMs increases hand artifacts and lip jitter; CSIM/HKV drop
- With RAMs + temporal latents, hands steady and faces sharper; identity improves
- Adaptations: reenactment uses 2D skeleton; talking-head crops face and disables hand RAM

Thank You

Questions & Discussion